

Health & Safety

Standard Underground construction work



Reference

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B	21-05-2024	A.JANSSENS	I.LELIEVRE O.BRZEZINKI	L.KNOLL	Document title modified Modification of 1. Traditional excavation

The underground work standards presented in this document serve as a practical guide to define the means to be planned and deployed.

Tunnel fire preparedness and emergency response requirements are covered in the document Emergency measures in underground environments [BYTP-INF-H&S-2172].

The elements of this document complement those presented in the following repositories:

- Standard Lifting [BYTP-REF-H&S-2210]
- Standard Fall from height prevention [BYTP-REF-H&S-2211]
- Standard Machine / Pedestrian Interactions [BYTP-REF-H&S-2212]
- Standard Stability [BYTP-REF-H&S-2213]
- Standard LOTO [BYTP-REF-H&S-2232]
- Standard Emergency Management [BYTP-REF-H&S-2221]

The elements of this document also complement the elements presented in the following MOSE+:

- 5750 – Construction of umbrella arch with short feeds
- 5755 – Explosive Drilling Charging Scaling of a blast
- 5756 – Concreting tunnel linings
- 5760 - Installation of ring with segment erector inside a TBM
- 5761 –Injection of backfill grout into outer surface of segments

Hyperbaric diving activities are not covered in this document and must be conducted by countries and projects in accordance with local regulations which cover organizational, technical and human requirements.

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1. Traditional excavation

The excavation method selection is directly linked to the geological, hydrogeological and environmental conditions on the project.

For mechanical excavation, the first selection criterion is whether or not the machine can support the ground.

Traditional excavation is carried out using various approaches:

- Mechanical excavation using an excavator
- Excavation by point attack machine (cutting, rock breaker, etc.)
- Use of explosives (Drill & blast)
- etc.



Checking the condition and evolution of the tunnel face and the vault is essential. Stopping points must be performed to allow the proper verification of their conditions.

Human presence at the tunnel face is only allowed at the authorization of the foreman/superintendent.

Attack Point Machines

For friable rock, the excavation is carried out via a machine which attacks the ground gradually and on a localised area.

These machines on tires or tracks are equipped with adjustable arms, at the end of which attack devices are installed (rock breaker, roadheader, etc.).

The walls are equipped with temporary support as it advances.



Use of explosives (Drill & Blast):

Drill & Blast operations are carried out by competent and qualified personnel, according to local regulations. For example :

- Firing attendant / Mining manager: Person competent in mining, prepares the firing plan, orders the explosives
- Shotfirer (Blaster): Responsible for the safety of mining operations, establishes the firing plan (if trained for it), carries out and monitors the explosives loading operations, has the Firing Officer Certificate (depending on local regulations), holds a shooting permit, deliver the access to the fire area, etc.



The Site Installation Plan includes the pyrotechnic installation, if required.

The safety distance between the attack point and the tunnel installations (ventilation, electricity, water, drainage, machinery, etc.). is defined according to the blast and projections generated by the explosions.

During the first blasts, install adequate protections to avoid the projections towards the outside of the tunnel (curtains, walls, etc.).

Strictly follow the procedures for starting and stopping ventilation during standard activities and during Blasts.



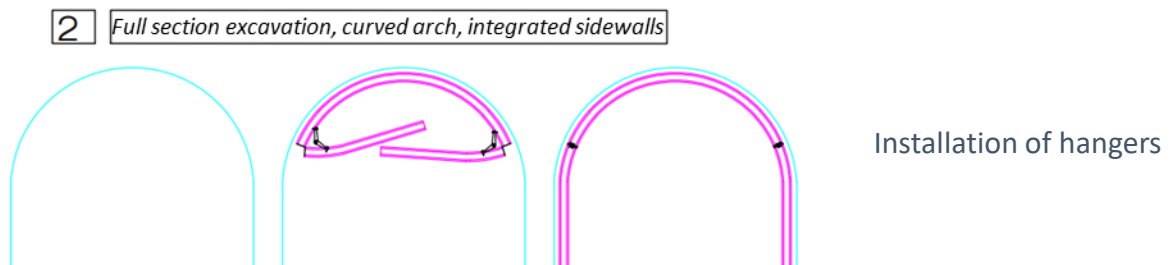
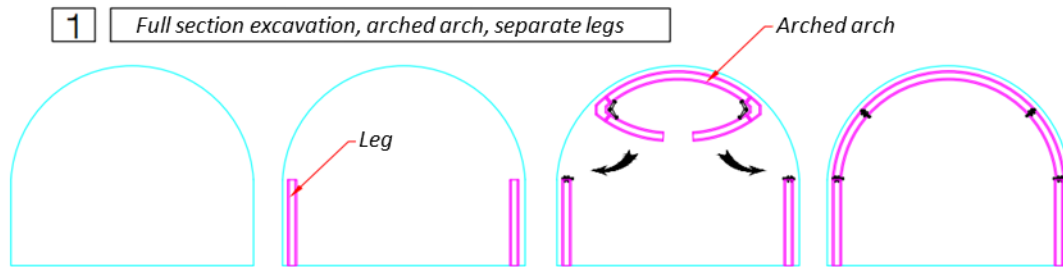
Access to ground withdrawal (scaling and mucking out works) areas, carried out by earthmoving engines, is strictly prohibited for pedestrians.

2. Temporary support

The temporary support is used to guarantee a safe stability of the vault during excavations of the tunnel face.

The type of support can be of various natures, by :

- Anchor bolts for homogenizing the behavior of the ground and/or supporting deformations
- Hangers, composed of steel elements, forming an arc or a gantry depending on the shape of the excavation. They are arranged according to the cross section of the tunnel, at regular intervals, along the excavated walls in order to absorb the geostatic pressures. There are 3 main types of hangers: sliding, reticulated and heavy.
- Setting up welded mesh
- Shotcrete (wet, semi-wet or dry) by hand or robot to complete the system to protect the surface of the excavation from alteration and to take up part of the loads from the ground



Shotcrete



Anchor bolts

3. Vault formwork

Two formwork methods (depending on the constraints and specificities of the site):

- Specific formwork, composed of a formwork skin and a framework - For vault formwork, these are placed on shoring (integrated into the formwork or independent, depending on the case)
- Use of a self-supporting metal formwork tool. It can be composed of a translation gantry, with or without skirt locking. The adjustment of this formwork is carried out by jacks and ball joints.



Specific formwork



Self-supporting metal formwork

Safety In Design:

It is required to involve the Safety Department early on for the design of the formwork, in order to ensure the integration of the intrinsic safety of the formwork and the workers' safety during the assembly/dismantling/use phases.

- Intrinsic safety
 - Access
 - Collective protections
 - Working heights
 - Lifting points
 - etc.
- Assembly and disassembly phases:
 - Assembly with portable power tools
 - Lifting the constituent parts with cranes or any lifting equipments
 - Temporary stability
 - Manual handling
 - Supply of equipment in the upper part (lock barrier, dimensioned jib, etc.)
- Use :
 - Bending of the formwork (hydraulic cylinders, etc.)
 - Installation of electrical cabinets on the formwork (avoids the length of extension cords)
 - Concrete supply lines
 - events
 - Integrated vibration
 - Water point available at the working place
 - Cleaning pipes with water (no compressed air)

The gantry for carrying out the waterproofing and reinforcement takes into account the same design elements as the main formwork.

Portal
sealing/reinforcement



4. Tunneling boring machine (TBM)

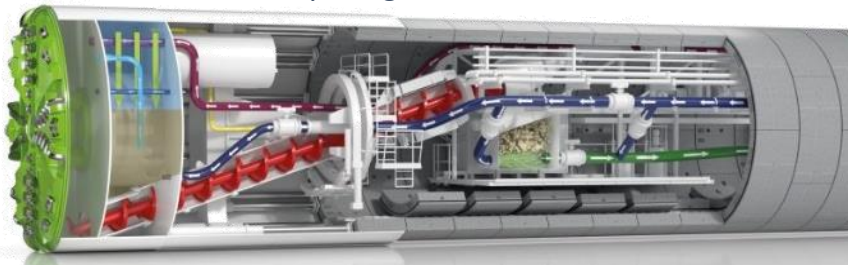
The tunnel boring machine performs the functions of excavation with lateral and frontal support.

It is provided at the front with a rotating cutting shield, which can be isolated from the rear of the tunnel by a watertight partition and in which a containment pressure is maintained intended to actively support the excavation and/or or balance the hydrostatic head developed by the ground water. However, some TBMs can dig in "open mode", without containment pressure exerted at the front.



Excavation is carried out by the cutterhead. The extraction of materials is carried out either by a screw or by pickling pumps.

Progression is ensured by thrust system consisting of thrust cylinders pushing against the tunnel segments installed under the shelter by using an erector.



Standard TBM scheme

Safety In Design:

Contact the H&S Department during the TBM design phase (access, collective protection, working heights, lifting points, weight of elements, etc.)

- Assembly :
 - Provide access to work at height around the entry joint, shield, cutting wheel, etc.
 - Provide additional ventilation for welding operations when necessary
- Commissioning : the TBM is at least compliant with the Machinery Directive, and takes into account the specificities linked to local regulations.



Inlet seal



Thrust frame



Welding cutterhead



- Factory assembly of pre-made segments:
 - Take into account working heights, weight of loads, etc.
 - The circulation plan of the factory is such that reversing steps are not necessary



5. External / surface logistics

Logistics is a major issue in tunnel productivity, and contributes significantly to better safety when this is understood from the design phase.

Traffic plan / Organization of supplies

The circulation plan for the tunnel supply/removal zones includes all the volumes handled (non-exhaustive list):

- Segments
- Hangers / welded mesh
- Extension elements for pedestrian bridges and pipelines
- Concrete, mortar, bentonite
- Grave cement, etc.
- Traditional branch formwork
- Operating equipment for digging machines (TBM, traditional)
- Scaling and mucking out works, conveyors
- Waste, etc.



Provide loading/unloading areas at fixed points with equipment allowing access and long-term collective protection, with clearly identified site entrance and exit.

Lifting equipments/moving elements

The elements handled are packaged in baskets/monolithic loads and the means of gripping and moving loads comply with local regulations and our lifting standards.



Lifting gantry



MSV (Multi Service Vehicle)



Chameleon excavator
(excavation of debris)

Surface storage / remote storage

- The platforms are sized for the weight of stored items
- Pedestrian paths are protected
- Truck rotations, arrivals and exits of materials are managed and anticipated by the platform manager



Evacuation of mucking
out by river



Storage of
segments

6. Tunnel access / access controls

Tunnel access

For access with negative height differences (identical to shafts):

- Set up a staircase tower or Escalib
- Install an elevator as soon as the negative height difference reaches 14m (starting at the excavation phase)
- The use of a crane platform is subject to a specific procedure (Refer to Standard Fall from height)

The installation of an elevator is mandatory whenever hyperbaric dives are planned.



Escalib



Elevator

Access controls in the “tunnel” zone

The workers accessing the “tunnel” zone are :

- Trained and qualified in tunnel-specific safety instructions
- Trained in the use of the self-rescue mask
- Equipped with an access control badge (RFID chip or equivalent)

Visitors accessing the “tunnel” zone are:

- Accompanied at all times by project staff (themselves trained in specific tunnel instructions)
- Informed of safety instructions specific to the tunnel and self-rescuer mask
- Equipped with an access control badge (RFID chip or equivalent)



RFID
chip



Self-rescuer
mask

The number of people present in the tunnel zone is displayed at the salient points (tunnel entrance, branches if necessary, TBM back up, etc.) and corresponds to the capacity of the refuge chamber(s) installed in the tunnel



7. Pedestrian traffic in tunnels

Pedestrian paths are physically separated from vehicle/construction equipment access routes:

- Installation of a tunnel walkway – Periodic control by qualified person to be carried out.
- Mark the parking areas for tunnel machines
- Mark out occasional intervention areas in the tunnel (topographical surveys, equipment maintenance, etc.)

Absence of tunnel walkway must be justified by an in-depth risk analysis and the decision validated by the H&S Department and the Project Manager

For mechanized staff travel:

- Use of MSV
- Construction vans equipped with all safety elements
- Light vehicles such as buggies for support staff/supervision



Tunnel walkway

8. Ventilation / management of air pollutants

There are multiple sources of air pollution in tunnel :

- Ground that can naturally emit gases
- Use of thermal construction machinery
- Use of chemicals
- Silica dust (concrete)
- ...

Air quality in tunnels must be compatible with the health and safety of workers (threshold local regulations on values depending on the pollutants must be respected).

Mechanical ventilation must be sized according to the length of the tunnel, as well as the openings created (cross-passages, stations, etc.), and justified by calculation note.

The air inlet source is located away from any source of pollution (vigilance in the event of urban pollution), and ensures a minimum flow rate of 25L/s/man at the working face (this data may vary depending on the digging method and the local regulations).



Dust management

- Limit the use of materials and equipment that generate dust
- Capture at source when possible
- Water traffic lanes when necessary
- Cover the conveyors if necessary

Gas management

- Favor anti-pollution electrical or thermal equipments
- Provide an automatic extinguishing system on the equipment



Anti-pollution filter



Automatic extinguishing on machine

It is strictly forbidden to charge electric construction equipment in tunnels (use the shafts or outside locations)

Dust and gas measurements

2 possible types of monitoring:

- By spot sampling and sending to the analysis lab
- By continuous monitoring

Prioritize continuous monitoring, as it provide the concentrations of previously defined gases and dusts in real time. The stations are connected to the network and the data is transferred and available automatically to a secure platform, supplemented if necessary, by individual measurements over 8 hours, sent to the laboratory



EcomSmart Station (ECOMESURE)

9. Lengthening of utility pipes and electricity

The extension of utility pipes is subject to specific procedures, in compliance with the site's electrical and mechanical LOTO procedures.

Example of color coding

Electricity and light	Red	Red
Fuel gas (transport/distribution), hydrocarbons	Yellow	Yellow
Chemical products	Orange	Orange
Potable water	Blue	Blue
Sanitation and Pluvial water	Brown	Brown
HVAC	Purple	Purple
Telecom	Green	Green
Traffic lights and road signs	White	White
Multi-network coverage area	Pink	Pink

- Pipes and networks must be identified
- For the supply of pipes, provide racks and suitable lifting equipment

Pipes extension on TBM

- Pipe extension systems (reels) must be protected and made inaccessible by enclosure.
- No fixed workstation near the reels

Reel protections



TBM pipe rack

10. Equipment maintenance

Each equipment maintenance is planned and anticipated.

- Mortar plant, mortar lines
- Ventilation
- TBM equipments

All maintenance interventions are recorded in a computerized maintenance management system

Provide at least a mechanical workshop area at the surface, and preferably also one of the TBM trailers.

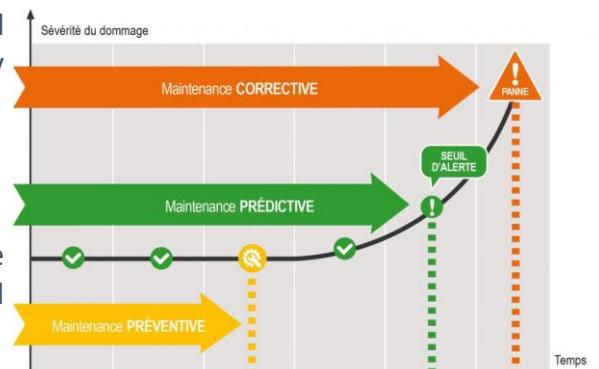
Implement maintenance programs based on the related equipment (manufacturer recommendations, regulatory checks, etc.):

- Predictive maintenance
- Preventive maintenance
- Corrective maintenance

Preventive maintenance maintains equipment while predictive monitors trends to predict breakdowns and prevent them

PROCEDURE : « PLAN DE MAINTENANCE HEBDOMADAIRE »	
Instructions	Illustration
- Vérifier la fixation des électrovannes	

Example of maintenance plan



A machine that is constantly maintained is much more productive and safer!

11. Emergency management

The Tunnel Emergency Response Plan is prepared before the beginning of any underground activity. Emergency exercises are carried out as many times as necessary, in collaboration with external Emergency Services when relevant.

It takes into consideration, among other things (non-exhaustive list):

- Access and access controls
- Evacuation routes
- Emergency equipment (survival shelter, self-rescue masks, etc.)
- Emergency lighting and tunnel markers
- The opening of the branches as the work progresses (if applicable)
- Communication means
- Fire detection and alarm systems
- Hydraulic resources
- ...



Details related to emergency management can be found in the Standard Emergency Management

Hydraulic resources

Emergency cabinet